

DR. GORDON GUOCHENG QIAN

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BIO

Senior Research Scientist at Snap Inc. with hands-on R&D experience in multimodal generation across image, video, and 3D. Member of the video generation team, working on pretraining and post-training for 5B to 30B VLM-diffusion models and currently focused on self-forcing for streamable video generation. Previously served as tech lead for personalized generation and built end-to-end systems spanning large-scale data curation, model post-training, model evaluation and cross-functional leaderboards, and product deployment. Authored **20+ top-tier publications (3.5K+ citations)**, filed **6 patents**, and shipped **3 research innovations** to Snapchat products including Dreams, EasyLens, and AI Lens, impacting **900M+** monthly active users. Serves as **Area Chairs for ICLR 2026 and NeurIPS 2026**.

PROFESSIONAL EXPERIENCE

Snap Inc.

Palo Alto, CA, USA

Senior Research Scientist

Dec 2023 – Present

- Focus: **Media generation (video, image, 3D) and personalization**.

- Member of the video generation team, contributing to both pretraining and post-training. Studied text-encoder and architecture choices for video generation, owned camera annotation for camera-control video generation projects (Wonderland, AC3D, VD3D), mentored video editing work EasyV2V, and video alignment tuning Diffusion-DRF (under review). Currently working on self-forcing for streamable video generation while mentoring agentic video generation projects;

- Tech lead of full-stack R&D for personalized image generation from data, model training, eval, to deployment. Shipped 3 products, demonstrated 3 prototypes, filed 6 patents, published 13 papers, received 4 outperformance awards, and earned 1 promotion.

HIGHLIGHTED PROJECTS

Multimodal Image Generation, Personalization, Reference-Based Generation

- Tech lead. Led multi-million-sample data collection, 10B+ model post-training, and leaderboard curation and benchmarking. Shipped 3 products and filed 3 patents. Highlighted: Canvas-to-Image (SIGGRAPH), Omni-ID (CVPR).

Video Camera Control, Editing, and Alignment Tuning

- Mentor. Collected datasets and annotated camera controls, contributed core ideas, and filed 2 patents. Highlighted: video editing EasyV2V (CVPR), video alignment tuning Diffusion-DRF (ECCV under review), and camera-controlled video generation with Wonderland (CVPR) and AC3D (CVPR).

Vision-Language Models (VLMs)

- Mentored projects across two agentic generation efforts (WIP). Representative work includes VLM post-training via Omni-Attributes (CVPR) and Think-Diff (ICML), VLM-Diffusion Canvas-to-Image (SIGGRAPH), and VLM-based reward tuning in Diffusion-DRF.

Image-to-3D Generation

- Led Magic123 (ICLR 2024, 450+ citations, 1.6K+ GitHub stars).

Scalable Spatial Understanding, 3D Representation Learning

- First-authored multiple projects, including PointNeXt (NeurIPS 2022, 34% accuracy gain, 1,200+ citations, 1K+ GitHub stars) and ASSANet (NeurIPS 2021 Spotlight, 54× speed-up, 120+ citations).

INTERNSHIP EXPERIENCE

April 20, 2026

Guocheng Qian · Résumé

<https://guochengqian.github.io/>

Snap Inc.

May 2023 – Sep 2023

- Project: Image/text-to-3D generation

Meta Platforms

Aug 2022 – Jan 2023

- Project: Efficient backbone for long-range video understanding

Microsoft Research Asia

Dec 2021 – May 2022

- Project: Scalable 3D geometric representation learning

Megvii Technology Research

Jul 2021 – Nov 2021

- Project: Tree-search based efficient neural architecture search

SenseTime Research

Aug 2018 – Apr 2019

- Project: Unified deep pipeline for demosaicing, denoising, and super-resolution

EDUCATION**King Abdullah University of Science and Technology (KAUST)**

Jeddah, Saudi Arabia

Ph.D. in Computer Science, specializing in 3D Vision & Generation

May 2019 - Dec 2023

*CEMSE Research Excellence Award (Top 12 among all 450 CEMSE students)***Xi'an Jiaotong University (XJTU)**

Xi'an, China

Bachelor's degree in Engineering, with honors

Aug 2014 - Jul 2018

*Outstanding Undergraduate (Top 10 among all 2000 undergraduates)***PUBLICATIONS**See Google Scholar for the full list: <https://scholar.google.com/citations?user=DUDaxg4AAAAJ>As of April 2, 2026: **3,572 citations**, h-index 19, **24 top-tier conference/journal papers**, and 9 co-first or corresponding author papers. Selected publications are listed below. Here, * denotes co-first authorship, and † denotes corresponding author.

- [1] Yusuf Dalva*, **Guocheng Qian***[†], Maya Goldenberg, Tsai-Shien Chen, Kfir Aberman, Sergey Tulyakov, Pinar Yanardag, and Kuan-Chieh Jackson Wang. Canvas-to-Image: Compositional image generation with multimodal controls. In *SIGGRAPH Conference Papers*, 2026 [paper] [project]
- [2] **Guocheng Qian**, Kuan-Chieh Wang, Or Patashnik, Negin Heravi, Daniil Ostashev, Sergey Tulyakov, Daniel Cohen-Or, and Kfir Aberman. Omni-ID: Holistic identity representation designed for generative tasks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025 [paper] [project]
- [3] **Guocheng Qian**, Daniil Ostashev, Egor Nemchinov, Sergey Tulyakov, Kuan-Chieh Jackson Wang, and Kfir Aberman. ComposeMe: Attribute-specific image prompts for controllable human image generation. *SIGGRAPH Asia*, 2025 [paper] [project]
- [4] Yifan Wang, Yanyu Li, **Guocheng Qian**[†], Sergey Tulyakov, Yun Fu, and Anil Kag. Diffusion-drif: Free, rich, and differentiable reward for video diffusion fine-tuning. *under review*, 2026 [paper] [project]
- [5] Jinjie Mai, Chaoyang Wang, **Guocheng Qian**, Willi Menapace, Sergey Tulyakov, Bernard Ghanem, Peter Wonka, and Ashkan Mirzaei. Easyv2v: A high-quality instruction-based video editing framework. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026 [paper] [project]
- [6] Anujraaj Argo Goyal, **Guocheng Qian**[†], Huseyin Coskun, Aarush Gupta, Himmy Tam, Daniil Ostashev, Ju Hu, Dhritiman Sagar, Sergey Tulyakov, Kfir Aberman, and Kuan-Chieh Jackson Wang[†]. Preventing shortcuts in adapter training via providing the shortcuts. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025 [paper] [project]
- [7] **Guocheng Qian**, Jinjie Mai, Abdullah Hamdi, Jian Ren, Aliaksandr Siarohin, Bing Li, Hsin-Ying Lee, Ivan Skorokhodov, Peter Wonka, Sergey Tulyakov, and Bernard Ghanem. Magic123: One image to high-quality 3d object generation using diffusion priors. In *International Conference on Learning Representations (ICLR)*, 2024 [paper] [project]  450,  1.6k

- [8] Tsai-Shien Chen, Aliaksandr Siarohin, **Guocheng Qian**, Kuan-Chieh Jackson Wang, Egor Nemchinov, Moayed Haji-Ali, Riza Alp Guler, Willi Menapace, Ivan Skorokhodov, Anil Kag, et al. Omni-attribute: Open-vocabulary attribute encoder for visual concept personalization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026 [paper] [project]
- [9] **Guocheng Qian**, Yuchen Li, Houwen Peng, Jinjie Mai, Hasan Hammoud, Mohamed Elhoseiny, and Bernard Ghanem. PointNeXt: Revisiting pointnet++ with improved training and scaling strategies. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 23192–23204, 2022 [paper]  Cites 1289  Stars 1k
- [10] **Guocheng Qian**, Hasan Hammoud, Guohao Li, Ali Thabet, and Bernard Ghanem. ASSANet: An anisotropic separable set abstraction for efficient point cloud representation learning. In *Advances in Neural Information Processing Systems (NeurIPS Spotlight)*, volume 34, pages 28119–28130, 2021 [paper]  Cites 126
- [11] **Guocheng Qian**, Abdulellah Abualshour, Guohao Li, Ali Thabet, and Bernard Ghanem. PU-GCN: Point cloud upsampling using graph convolutional networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 11683–11692, June 2021 [paper]  Cites 349  Stars 199  Cites 126
- [12] **Guocheng Qian**, Xingdi Zhang, Abdullah Hamdi, and Bernard Ghanem. Pix4Point: Image pretrained transformers for 3d point cloud understanding. In *International Conference on 3D Vision (3DV)*, 2024 [paper]
- [13] **Guocheng Qian**, Yuanhao Wang, Jinjin Gu, Chao Dong, Wolfgang Heidrich, Bernard Ghanem, and Jimmy S Ren. Rethinking learning-based demosaicing, denoising, and super-resolution pipeline. In *IEEE International Conference on Computational Photography (ICCP)*, pages 1–12. IEEE, 2022 [paper]  Stars 298
- [14] **Guocheng Qian**, Ruihang Zhang, Tsai-Shien Chen, Yusuf Dalva, Anujraaj Goyal, Willi Menapace, Ivan Skorokhodov, Daniil Ostashev, Meng Dong, Arpit Sahni, Ju Hu, Sergey Tulyakov, and Kuan-Chieh Jackson Wang. LayerComposer: Multi-human personalized generation via layered canvas. *Under review*, 2025 [paper] [project]
- [15] Thuan Hoang Nguyen, Jiahao Luo, Hao Li, Jian Wang[†], and **Guocheng Qian**[†]. FFAvatar: Few-shot, feedforward, and generalizable avatar reconstruction. *under review*, 2025
- [16] G. Li*, Matthias Müller*, **Guocheng Qian***, Itzel C. Delgadillo, Abdulellah Abualshour, Ali K. Thabet, and Bernard Ghanem. DeepGCNs: Making gcns go as deep as cnns. *IEEE transactions on pattern analysis and machine intelligence (T-PAMI)*, 45(6):6923–6939, 2023 [paper] [project]  Stars 629
- [17] G. Li*, **Guocheng Qian***, Itzel C. Delgadillo*, Matthias Müller, Ali K. Thabet, and Bernard Ghanem. SGAS: Sequential greedy architecture search. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1617–1627, 2020 [paper] [project]  Cites 290  Stars 162
- [18] Hanwen Liang, Junli Cao, Vidit Goel, **Guocheng Qian**, Sergei Korolev, Demetri Terzopoulos, Konstantinos Plataniotis, Sergey Tulyakov, and Jian Ren. WonderLand: Navigating 3d scenes from a single image. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025 [paper] [project]
- [19] Sherwin Bahmani, Ivan Skorokhodov, **Guocheng Qian**, Aliaksandr Siarohin, Willi Menapace, Andrea Tagliasacchi, David B. Lindell, and Sergey Tulyakov. AC3D: Analyzing and improving 3d **Camera Control in Video Diffusion** transformers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025 [paper] [project]
- [20] Sherwin Bahmani, Ivan Skorokhodov, Aliaksandr Siarohin, Willi Menapace, **Guocheng Qian**, Michael Vasilkovsky, Hsin-Ying Lee, Chaoyang Wang, Jiaxu Zou, Andrea Tagliasacchi, David B. Lindell, and Sergey Tulyakov. VD3D: Taming large video diffusion transformers for 3d camera control. In *International Conference on Learning Representations (ICLR)*, 2025 [paper] [project]
- [21] Zhenxing Mi, Kuan-Chieh Wang, **Guocheng Qian**, Hanrong Ye, Runtao Liu, Sergey Tulyakov, Kfir Aberman, and Dan Xu. I think, therefore i diffuse: Enabling multimodal in-context reasoning in diffusion models. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2025 [paper] [project]
- [22] Tuan Duc Ngo, Ashkan Mirzaei, **Guocheng Qian**, Hanwen Liang, Chuang Gan, Evangelos Kalogerakis, Peter Wonka, and Chaoyang Wang. DELTA_{v2}: Accelerating dense 3d tracking. In *Under review*, 2025 [paper] [project]

- [23] Abdullah Hamdi, Luke Melas-Kyriazi, Jinjie Mai, **Guocheng Qian**, Ruoshi Liu, Carl Vondrick, Bernard Ghanem, and Andrea Vedaldi. GES: Generalized exponential splatting for efficient radiance field rendering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024 [paper]
- [24] Chen Zhao, Shuming Liu, Karttikeya Mangalam, **Guocheng Qian**, Fatimah Zohra, Abdulmohsen Alghannam, Jitendra Malik, and Bernard Ghanem. Dr2Net: Dynamic reversible dual-residual networks for memory-efficient finetuning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024 [paper]
- [25] Jinjie Mai, Wenxuan Zhu, Sara Rojas Martinez, Jesus Zarzar, Abdullah Hamdi, **Guocheng Qian**, Bing Li, Silvio Giancola, and Bernard Ghanem. Tracknerf: Bundle adjusting nerf from sparse and noisy views via feature tracks. In *Proceedings of the European Conference on Computer Vision (ECCV)*, 2024 [paper] [project]
- [26] Tianyu Zhang, **Guocheng Qian**, Jin Xie, and Jian Yang. Fastpci: Motion-structure guided fast point cloud frame interpolation. In *Proceedings of the European Conference on Computer Vision (ECCV)*, 2024 [paper]
- [27] Guohao Li, Chenxin Xiong, **Guocheng Qian**, Ali Thabet, and Bernard Ghanem. DeeperGCN: Training deeper gcns with generalized aggregation functions. *IEEE transactions on pattern analysis and machine intelligence (T-PAMI)*, 2023 [project]  Stars 629
- [28] Jun Chen, Deyao Zhu, **Guocheng Qian**, Bernard Ghanem, Zhicheng Yan, Chenchen Zhu, Fanyi Xiao, Mohamed Elhoseiny, and Sean Chang Culatana. Exploring open-vocabulary semantic segmentation without human labels. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023 [paper]

ACADEMIC EXPERIENCE

Teaching Assistant Deep Learning for Visual Computing (2020 - 2022), Design and Analysis of Algorithms (2021)

Area Chair ICLR 2026, NeurIPS 2026

Reviewer CVPR, ECCV, ICCV, ICML, NeurIPS, SIGGRAPH, SIGGRAPH Asia, AAAI, IJCAI, T-PAMI, TVCG

Conference Organizer Tutorial on Graph Machine Learning for Computer Vision @ CVPR'22 , Workshop on Learning 3D with Multi-View Supervision @ CVPR'23, CVPR'24, and CVPR'26

HONORS & AWARDS

KAUST CEMSE Research Excellence Award Top 12/450

KAUST CEMSE Dean's List Award Top 20%

KAUST Fellowship Full tuition support, monthly stipend, housing, and medical coverage

National First-Class Scholarship Top 1% nationwide

Outstanding Undergraduate Award Top 10/2000

Excellent Student Cadre Top 3%

TECHNICAL SKILLS

- **Multimodal Representation, Understanding, and Generation:** 20+ top-tier publications across representation learning, spatial understanding, multimodal generation, and alignment fine-tuning
- **PyTorch & Open-source:** Magic123  >1.6K+ stars, PointNeXt  1K+ stars, DeepGCNs  1.2K+ stars
- **Large-scale Systems:** built end-to-end multimodal pipelines covering large-scale data collection, annotation, processing, and distributed training on up to 256 GPUs; developed data-visualization tools for quality control and benchmark leaderboards for model evaluation.
- **End-to-end R&D:** shipped 3 research innovations in Snapchat products, including Dreams, EasyLens, and AI Lens